ZetaERP — Manual Bug-Testing SOP

Part 9 — POS (Point of Sale): the billing terminal, sessions, payments, receipts, hold/recall

Document	QA-SOP-09 — POS
Product	ZetaERP / Zenith (multi-tenant POS / ERP)
Environment under test	Production — https://erp.zetaclantech.com
Scope of this part	The **POS** module (`/apps/zenith/pos/…`): the **Billing Terminal** (product grid, cart, barcode scan, qty/price edit, GST tax toggle, sales-mode switching, customer attach, **Cash/Card/UPI + Split** payments, receipt auto-print, New Sale, **Hold/Void/Recall**, **bill edit / amend**, **credit / pay-later**), **Sessions** (open/close + cash reconciliation), **Terminals / Sales Modes / Product Prices** config, **Receipt** templates, **bill numbering**, **GST** on the bill, the **POS→Sales-invoice sync**, **offline mode**, reports/audit, and **multi-tenant isolation**.
Audience	Manual QA tester (no code knowledge assumed; basic GST literacy helps)
Document owner	Platform / Solo-dev
Status	Active
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1. Purpose

This SOP tells a tester **exactly how to manually test the POS terminal A→Z**, including edge and UI cases: open a session, ring up a sale (scan/tap products, edit qty/price, switch sales mode, toggle GST, attach a customer), take payment (cash/card/UPI and **split**), print the receipt, start a new sale, and **hold/recall/void/amend** bills — then verify the **money and GST are correct**, the **bill number is right and gap-free**, the **stock and Sales invoice sync**, and everything is **isolated per tenant**.

> **The POS is the live cash register.** A wrong total, a double-charge, a mis-numbered bill, or a cross-tenant leak is a real money/compliance problem. Do the arithmetic by hand, and **never test on MARUTHI's (or any live tenant's) real terminal, sessions, or bills.**

2. Critical rules for the tester (read before starting)

> ⚠️ **This is a LIVE cash register for a real shop (MARUTHI, org `100000`).**

1. **NEVER open a session, ring a sale, take a payment, void, hold, amend, or close a session** on the live flagship tenant **MARUTHI** or any live tenant's terminal. Use a **throwaway tenant** with its own terminal.
2. **Never alter or renumber an already-issued bill.** Editing a bill (amend) keeps the same number by design — do that only on throwaway bills.
3. **Never take a real payment.** Cash/Card/UPI here record against a test session; don't run a real card or UPI. The terminal does **not** hit a live payment gateway (payments are recorded, not charged), but still only use a throwaway session.
4. **Multi-tenant isolation is a top surface** (§14). Products, transactions, sessions, and terminals must be scoped to the active org; any cross-tenant read/write is **Critical**.
5. **GST on the bill is a legal record** — verify the CGST/SGST vs IGST split and the total by hand (§8). The POS→Sales-invoice sync makes the bill a tax invoice.
6. **Clean up** every test session/bill/terminal/product (and any throwaway tenant), or hand the dev the list. Don't leave an open test session or a PtCreated cart behind.
7. **Record the exact steps + amounts** for every case; for a money/GST case a bare "looks right" is not acceptable.

3. Test environment & preconditions

The POS is at `/apps/zenith/pos/<screen>` (lowercase `pos`; the checkout is capital-**T** `Terminal`). It renders on **/apps (Bootstrap)**; the rich optimistic terminal is **/apps-only** — **/qapps strips `<script>` / `<style>`**, so the Terminal client-side bounces `/qapps → /apps`. Test on **/apps**.

Screen (URL)	Purpose
`/apps/zenith/pos` (→ Dashboard)	POS home / KPIs
`/apps/zenith/pos/Terminal`	**The billing terminal (checkout)**
`/apps/zenith/pos/PosSessions` (+ EditSession)	Open/close sessions, cash
`/apps/zenith/pos/PosTerminals`	Terminals config
`/apps/zenith/pos/PosTransactions` (+ EditTransaction)	Bill list (EditTransaction is legacy back-office — see §16)
`/apps/zenith/pos/CreditBills`	Pay-later bills
`/apps/zenith/pos/SalesMode` · `ProductPricing`	Sales modes · per-mode product prices
`/apps/zenith/pos/ReceiptConfig`	Receipt templates
`/apps/zenith/pos/Reports` · `BillNumberRegister` · `BillAuditReport`	Reports · bill register · amendment audit
`/apps/zenith/pos/OfflineMode`	PWA offline config + sync

Set up on the throwaway tenant: a **terminal** (PosTerminals), a few **products** (some priced for Retail/Wholesale/Employee via Product Prices, some barcoded), a customer with a **GSTIN in the tenant's state** (intra) and one in **another state** (inter), the tenant's **GST config** (so the split works), and a **receipt template**. Optionally a real **barcode scanner** (it just types digits + Enter).

Browsers: full pass on Desktop Chrome; spot-check Edge/Safari/mobile/tablet. **Tools:** a **calculator** (verify GST + change), a **barcode scanner** (or keyboard), a phone for UPI/QR display checks, DevTools (Console/Network), and (optionally) **QZ Tray** for silent printing.

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4. Test data

Label	Use
Throwaway tenant + terminal	All POS testing
`PROD_18` / `PROD_5`	Products @18% / @5% GST with HSN + a scan code
`PROD_WS`	Product with a Wholesale price + a quantity-break threshold
`PROD_NOWS`	Product with **no** Wholesale price (should block in Wholesale mode)
`CUST_SAME` / `CUST_OTHER`	Customers with GSTIN in the tenant's / another state
Amounts	1 × ₹100 @18% → taxable 100, GST 18 (intra CGST 9 + SGST 9), total 118

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5. Conventions & severity

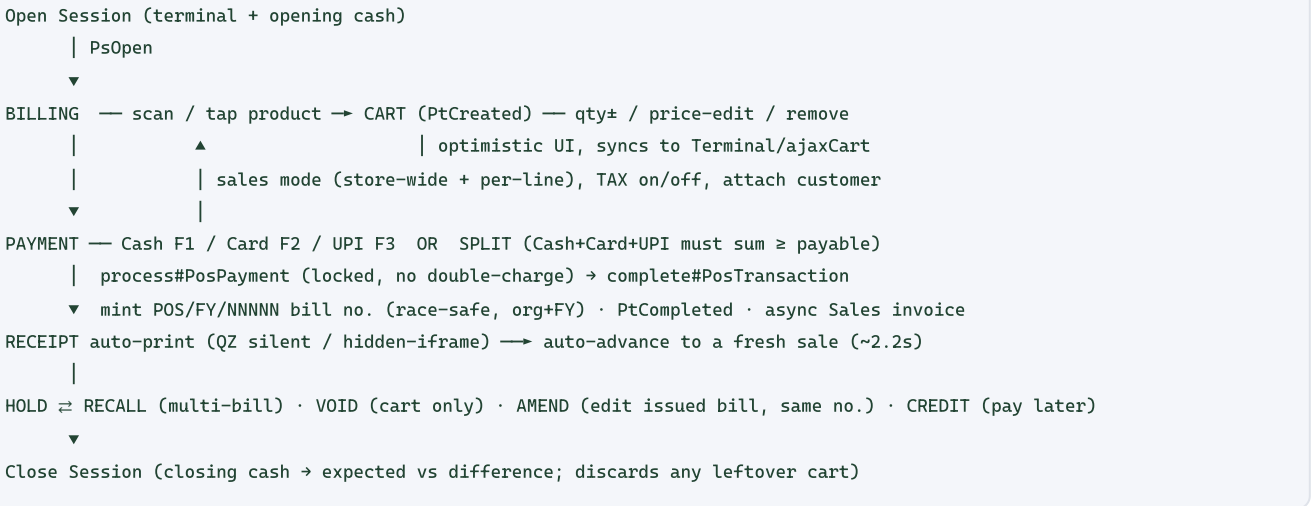
Case Type: `FUNC` · `GST` · `PAY` (payment) · `LIFE` (session/lifecycle) · `SCAN` · `EDGE` · `UI` · `A11Y` · `SEC` · `ISO` · `PERF`.

Sev	Meaning	Example
S1 — Critical	Wrong total/change; double-charge; wrong or duplicate bill number; wrong GST on the issued bill; cross-tenant product/txn/session access; a completed sale lost	split total doesn't equal cash taken; two bills share a number; IGST on an intra-state sale
S2 — High	A checkout flow broken or clearly wrong	qty edit deletes the line; unpriced product billable in a fixed mode; receipt never prints; session won't close
S3 — Medium	Wrong/loose behaviour, limited impact	quantity-break doesn't trigger; tax starts in the wrong state on load; round-off off by a paisa
S4 — Low	Cosmetic / minor UX	grid tile spacing; mode-button colour; receipt alignment

Logging a defect (template §16): Case ID, exact steps, **amounts (by-hand expected vs actual)**, tenant + terminal, session id, bill number, browser, screenshot (incl. the receipt), Console/Network. Evidence **mandatory** for GST, PAY, ISO, SEC.

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6. Flow map (a sale, A→Z)



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7. THE TERMINAL — checkout A→Z

Screen: `/apps/zenith/pos/Terminal`. Sections: **No-Session** (open a session) → **Billing** (cart/grid/payment) → **Receipt** (post-sale "Sale Complete!" then auto-New-Sale).

7.1 Session + cart basics

ID	Type	Title	Steps	Expected result	Sev
PS-001	LIFE	Open a session	Pick a terminal + opening cash → Open	Session opens (PsOpen); terminal warmed; billing screen shows	S2
PS-002	LIFE	One open session per terminal	Open a 2nd session on a terminal that already has one **with sales**	Blocked; an empty prior session is auto-closed instead	S2
PS-003	FUNC	Tap a product	Tap a grid tile	Line added to cart instantly (optimistic); total updates; cart is a PtCreated txn	S1
PS-004	FUNC	Category tabs	Click a category tab, then "ALL ITEMS"	Grid filters to that category / shows all (org's products)	S3
PS-005	FUNC	Product search	Type a name/code in the grid search	Grid filters (org-scoped)	S3
PS-006	FUNC	Qty + / −	Use +/− on a line	Qty and line total recompute; tax recomputes	S2
PS-007	EDGE	**Qty edit never deletes**	Set a line qty to 0 / empty / negative	Line is **not** deleted (reverts / floors); **only the × button removes**	S2
PS-008	FUNC	Price edit	Type a new unit price on a line	Line reprices; logged as a price override; line marked manual-price	S2
PS-009	EDGE	Negative price	Type a negative price	Rejected	S2
PS-010	FUNC	Remove line (×)	Click × on a line	Line removed; totals recompute	S3
PS-011	EDGE	Empty-cart checkout	With an empty cart	Payment panel hidden / cannot complete a zero-item sale	S2

7.2 Barcode scan

ID	Type	Title	Steps	Expected result	Sev
PS-015	SCAN	Scan a coded product	Scan into the cart search box	Correct product added instantly	S1
PS-016	SCAN	Unknown barcode	Scan a code with no product	Clear "Product not found"; nothing added	S2
PS-017	SCAN	Duplicate/ambiguous code	A code matching >1 product	Uses `list()` not `one()` → never silently mis-bills; resolves sensibly (see Part 7 Barcode)	S1
PS-018	SCAN	Scanner Enter	Real handheld (types digits + Enter)	Enter adds; no stray newline	S2

7.3 Sales modes, tax toggle, customer

ID	Type	Title	Steps	Expected result	Sev
PS-020	FUNC	Store-wide mode switch	Switch Retail → Wholesale (mode buttons)	Whole cart reprices to Wholesale; **Wholesale requires a customer** + a min order	S2
PS-021	FUNC	Per-line mode	Change one line's mode via its dropdown; use "↻ Auto" to reset	That line reprices; a manually-set mode shows 🔒 (locked); Auto reverts to Retail + re-enables quantity-break	S2

ID	Type	Title	Steps	Expected result	Sev
PS-022	EDGE	**Unpriced product in a fixed mode**	Add `PROD_NOWS` in Wholesale (no wholesale price)	**Blocked** at add and at mode switch (a clear "no price for this mode" message)	S2
PS-023	FUNC	**Quantity-break**	Add `PROD_WS` at Retail; increase qty past its threshold	Line auto-switches to the Wholesale price at the threshold; reverts below; **no-op for products with no break configured** (MARUTHI-safe); a manual-price line is exempt	S2
PS-024	GST	**TAX toggle OFF/ON**	Toggle GST OFF then ON	OFF → all line tax zeroed + totals drop; ON → GST recomputed per line (exclusive). **[Fixed 80dd4b0 — a fresh session now shows "TAX: OFF" consistently (the UI default matched to the billing logic's OFF default); the button no longer falsely shows "ON". Toggle still works both ways.]**	S3
PS-025	FUNC	Attach a customer	Customer button → search (name/phone) → select	Customer attached to the open bill (name/phone/GSTIN); org-scoped search	S2
PS-026	SEC	Customer cross-org	Try to attach a customer from another tenant	Rejected (same-org owner check)	S1

7.4 GST correctness on the sale (do the math)

POS calls the India GST engine (`calculate#Tax`) at add/reprice/mode-change/tax-toggle; prices are **tax-exclusive**. Intra vs inter comes from the terminal's origin region vs the customer's.

ID	Type	Title	Inputs	Expected result (by hand)	Sev
PS-030	GST	Intra-state 18%	`CUST_SAME` , 1 × ₹100 @18%	taxable 100; **CGST 9 + SGST 9**; total 118	S1
PS-031	GST	Inter-state 18%	`CUST_OTHER` , 1 × ₹100 @18%	taxable 100; **IGST 18**; total 118	S1
PS-032	GST	Mixed rates	₹100 @18% + ₹200 @5%	line1 GST 18, line2 GST 10; header tax 28; total 328	S1
PS-033	GST	Round-off	A total like ₹117.90	round-off applied to the payable (nearest step); receipt shows round-off	S2
PS-034	GST	Tax OFF sale	Toggle GST OFF, complete	Bill has 0 tax; the synced invoice is a **Bill of Supply** (tax 0), else a **Tax Invoice**	S2

7.5 Payments — Cash / Card / UPI + Split (do the math)

ID	Type	Title	Steps	Expected result	Sev
PS-040	PAY	Cash exact (F1)	Cash, amount = payable	Sale completes; change 0; bill number minted	S1
PS-041	PAY	Cash over-payment	Cash > payable	**Change = paid − payable** shown/printed; sale completes	S1
PS-042	PAY	Card (F2) / UPI (F3)	Pay by Card / UPI	Records a PosPayment of that method; completes	S1
PS-043	PAY	**Split payment**	SPLIT → e.g. ₹50 Cash + ₹68 UPI on a ₹118 bill	Each >0 amount records its own PosPayment; completes; **sum must be ≥ payable**	S1
PS-044	PAY	**Split under total (rejected)**	Split amounts sum **< payable**	**Rejected** — sale does not complete	S1
PS-045	PAY	**No double-charge on retry**	Double-click Pay / re-submit after completing	The payment path is **locked/idempotent** — no second charge, no second bill	S1
PS-046	PAY	Over-payment change on split	Split summing > payable	Change computed correctly; but **non-cash (Card/UPI/etc.) tenders may NOT exceed payable** — only CASH can over-tender (change is returned from cash). Card/UPI over the bill total is **rejected** (ZE-POS-0208)	S1
PS-047	PAY	**Split non-cash over-tender (rejected)**	Split where Card+UPI (non-cash) sum **> payable**, e.g. UPI ₹5,000 + Cash ₹3,000 on a ₹200 bill	**Rejected** at complete#PosTransaction ("Card/UPI/non-cash payments exceed the bill total...") — atomic, no bill number minted; prevents un-returnable "change" corrupting session cash reconciliation (ZE-POS-0208)	S1

ID	Type	Title	Steps	Expected result	Sev
PS-048	UI	**Split inputs legible in dark mode**	Open SPLIT (F4) on Safari/Mac with OS dark mode; type amounts, unfocus	Cash/Card/UPI amount + reference inputs stay legible (white field / dark text via explicit `color` + `background` + `color-scheme:light`), not dark-on-dark (ZE-POS-0209)	S3

7.6 Complete → receipt → new sale

ID	Type	Title	Steps	Expected result	Sev
PS-050	FUNC	Complete a sale	Pay	Status **PtCompleted**, `POS/FY/NNNNN` number minted at completion (not at cart start → no gaps); "Sale Complete!" summary	S1
PS-051	FUNC	**Receipt auto-prints**	After the sale	80 mm receipt prints via **QZ Tray silent** (if connected) or a **hidden-iframe** fallback; correct items/tax/total/change	S1
PS-052	FUNC	Receipt never blocks a sale	Force a receipt error (e.g. no template)	The sale stays completed; response flags `receiptUnavailable` — **the completed sale is never lost/undone**	S1
PS-053	FUNC	**Auto-advance to new sale**	Wait ~2.2s after the sale	Auto-advances to a fresh empty sale; a **pointer-down cancels** the auto-advance; the summary does **not** loop back	S2
PS-054	FUNC	New Sale / F10	Click New Sale (or F10)	Fresh empty cart; the previous completed bill is not carried forward	S2
PS-055	FUNC	Receipt PDF	Open a bill in **PosTransactions** → **EditTransaction** (back-office) and click **Download PDF**; also the post-sale summary's Download-PDF button	Downloads the server XSL-FO PDF (`/fop/apps/zenith/pos/ReceiptPdf?transactionId=…` , filename `Receipt-<id>.pdf`), same content as the printed receipt (items/qty/tax split/total/change/bill no.), using **this org's** receipt template (ZE-POS-0210)	S3

> **Note (ZE-POS-0210):** the ReceiptPdf XSL-FO screen existed but errored on every render (invalid FreeMarker `as number` ; missing `encode="false"` / `no-boundary-comment` ; missing `standalone="true"` so the POS chrome wrapped its `fo:root`). Fixed, org-scoped the template to the bill's org, and wired a reliable **Download PDF** in EditTransaction (via `/fop` , like sales InvoicePrint) plus repointed the post-sale Terminal buttons `/apps` → `/fop` . The reliable path is EditTransaction (the post-sale summary auto-advances ~2.2s).

7.7 Hold / Recall / Void

ID	Type	Title	Steps	Expected result	Sev
PS-060	LIFE	Hold a cart	Cart with items → HOLD (reason)	Cart snapshotted to a held bill; the terminal is freed for a new sale (transactionId cleared)	S2
PS-061	LIFE	Recall	RECALL → pick a held bill	The held items return; **recall first auto-holds the current non-empty cart** (no merge/loss)	S2
PS-062	LIFE	Multi-bill hold	Hold two carts, recall each	Both preserved; switching between them loses nothing (native `<details>` picker)	S2
PS-063	LIFE	Void	Void a cart	PtCreated → **PtVoided** (cart only); can't void a completed bill; the terminal is freed	S2
PS-064	EDGE	Void/hold then new bill	After hold/void, add a product	A **fresh** cart starts (not "transaction already completed/voided")	S2

7.8 Bill edit (amend) & credit / pay-later

ID	Type	Title	Steps	Expected result	Sev
PS-070	FUNC	**Amend an issued bill**	History → open an issued bill → edit items in the cart → re-bill	Re-bills the **same bill number**; stock adjusts by the **delta**; refund/balance reconciled; an **amendment audit** row is written	S1
PS-071	FUNC	Amend receipt reprint	After amend	The updated receipt reprints from the same receipt data	S2

ID	Type	Title	Steps	Expected result	Sev
PS-072	FUNC	Credit / pay-later	Complete a sale as **credit**	Bill created `isCredit='Y'`, paid 0; a per-phone customer party is created deterministically	S2
PS-072a	FUNC	**Credit marker on the receipt**	Complete a credit sale → auto-print; also reprint from History and Download PDF	Every receipt path shows **"** CREDIT / PAY LATER (UNPAID) **"** + **Amount Due <outstanding>** (ZE-POS-005). Covers all 4 builders: server `_rd` / `printThermalReceipt`, JS `buildReceiptHtml` + `buildHistReceipt`, and the XSL-FO `ReceiptPdf`. Non-credit bills show none of it	S2
PS-073	PAY	Settle a credit bill	Credit Bills → the page **loads the outstanding dues list** (each row = customer + due), then settle	Page loads dues (NOT stuck on "Loading credit bills…"); settle records the payment via `Terminal/ajaxCart action=settleCredit`; `isCredit` clears when fully paid (ZE-POS-002: the page needs a rendered `moquiSessionToken` or the ajaxCart POST 401s and the list hangs)	S2
PS-073a	FUNC	**Credit bill status in Transactions**	Transactions History → find a completed credit bill with a balance	Status reads **"Credit — Due <outstanding>"**, NOT "Completed" (the sale is `PtCompleted` but the receivable is open). Fully-settled/normal bills read "Completed"; voided read "Voided" (ZE-POS-004, display-only — underlying `statusId` unchanged)	S2

8. SESSIONS, BILL NUMBERING & INVOICE SYNC

8.1 Sessions & cash

ID	Type	Title	Steps	Expected result	Sev
PS-080	LIFE	Close session	Close with closing cash	`expected = opening + cash received − change given`; **cash difference** computed; leftover PtCreated cart discarded; idempotent (re-close is a no-op)	S2
PS-081	LIFE	Stale-cart cleanup	Open a cart, don't pay, re-open the session / start fresh	Abandoned PtCreated carts are retired (open/close/freshSale cleanup) — no old items reappear	S2
PS-082	LIFE	Cash entry / movement	Record a cash in/out	Reflected in the session cash reconciliation	S3

8.2 Bill numbering (race-safe, org-scoped)

ID	Type	Title	Steps	Expected result	Sev
PS-085	FUNC	Consecutive numbers	Complete several sales	`POS/FY/NNNNN` increments by 1, gap-free (minted at completion, not cart start)	S1
PS-086	PERF	**Concurrent terminals**	Two terminals complete sales at the same instant	**No duplicate/skipped** numbers (row-locked `PosBillSequence` per org+FY)	S1
PS-087	ISO	Per-tenant series	Compare numbers across two tenants	Each tenant has its own `org_FY` sequence — no cross-tenant sharing	S2

8.3 POS → Sales invoice sync

ID	Type	Title	Steps	Expected result	Sev
PS-090	FUNC	Bill becomes an invoice	Complete a sale → check Sales	A finalized Sales invoice is created (async), **idempotent** (one per bill via `linkedInvoiceId`), using the POS bill number as the tax-invoice number	S2
PS-091	GST	Invoice GST matches	Compare the invoice's GST to the bill	Same CGST/SGST/IGST breakdown carried to the invoice items; **Tax Invoice if tax>0 else Bill of Supply**	S1
PS-092	FUNC	Stock depletes	Sell an inventory-tracked product	Stock reduces on completion (only for `inventoryTracked='Y'` products)	S2

9. CONFIG — Terminals, Sales Modes, Product Prices, Receipts

ID	Type	Title	Steps	Expected result	Sev
PS-100	FUNC	Terminal CRUD	PosTerminals → create/edit a terminal	Persists with `organizationPartyId`, GSTIN, UPI VPA, state	S3
PS-101	FUNC	Sales modes	SalesMode → inspect Retail/Wholesale/Employee	Retail (MRP, default, exclusive); Wholesale/Employee (fixed price, require customer)	S3
PS-102	FUNC	Product Prices bulk grid	ProductPricing → set per-mode prices + a quantity-break	Persists per `(product, mode)`; a fixed-mode product with no price is blocked at the till	S2
PS-103	FUNC	Receipt template	ReceiptConfig → edit company/header/footer/width (58/80mm), default	Reflected on the printed receipt; discount limit / round-off step honoured	S3

10. OFFLINE MODE (PWA) & REPORTS

ID	Type	Title	Steps	Expected result	Sev
PS-110	FUNC	Offline config	OfflineMode → enable + export catalog	Catalog exported (**org-scoped**); PWA installable (manifest/sw)	S3
PS-111	EDGE	Offline sale sync	(Dev-assisted) queue an offline sale, sync	Idempotent via `OFFLINE:<id>` payment ref — **no double-post** on retry	S2
PS-112	FUNC	Reports	Reports → Bill-wise / Item-wise / Payment-wise / User-wise / Cancelled	Org-scoped totals reconcile to the session/bills	S3
PS-113	FUNC	Bill register / audit	BillNumberRegister; BillAuditReport	Consecutive numbers listed; amendment audit shows who changed what	S2

11. MULTI-TENANT ISOLATION (high priority)

`PosTransaction / PosSession / PosTerminal` (and PosExpense, PosCashEntry, price overrides, KOT, QR) are org-filtered (`MANTLE_ACTIVE_ORG`). The terminal scopes the grid, categories, customer search, catalog export, and scan by org.

ID	Type	Title	Steps	Expected result	Sev
ISO-001	ISO	Grid/transactions scoped	Cashier on tenant A	Only A's products, bills, sessions, terminals	S1
ISO-002	ISO	Same barcode two tenants	A and B each have a product with the same code	Each till resolves its own product	S1
ISO-003	SEC	Direct-URL to another tenant's bill/session	Open `EditTransaction` / `EditSession` with B's id	Blocked / not found	S1
ISO-004	ISO	Catalog export / offline sync	A exports catalog / syncs	Only A's products; sync can't post to B	S1
ISO-005	SEC	**Child entities via a direct screen**	Any screen listing `PosPayment` / `PosTransactionItem` / `PosHeldTransaction` / `PosProductModePrice` directly	These have **no direct org filter** (reached only via their org-scoped parent) — verify no direct-entity screen leaks them cross-tenant	S2

12. UI / ACCESSIBILITY / SECURITY

ID	Type	Title	Check	Expected result	Sev
PS-120	UI	Terminal on a real till screen	1366×768 / 1080p / a small 10" tablet	Grid + cart + payment fit; no clipping; big tap targets	S3
PS-121	UI	**/qapps bounces to /apps**	Open `/qapps/zenith/pos/Terminal`	Redirects to `/apps/...` (the rich terminal is /apps-only)	S2

ID	Type	Title	Check	Expected result	Sev
PS-121a	NAV	**Dashboard "Manage" shortcut tiles open**	Dashboard → click each Manage tile (Expenses, Cash Entry, Stock, Wastage, Kitchen, Import, Sales Mode, Receipts, Sessions, Terminals, Price Overrides, Credit Bills)	Each opens its screen (NOT a 404). Tiles link to the absolute `/apps/zenith/pos/<Screen>` — a raw `<a href="goX">` transition name resolves to `…/pos/goX` (sibling) → 404 on both surfaces (ZE-POS-003)	S2
PS-122	A11Y	Keyboard shortcuts	F1 (Cash) / F2 (Card) / F3 (UPI) / F10 (New Sale)	Fire the right action; visible focus	S3
PS-123	SEC	XSS in product/customer name	Product/customer name `<img src=x onerror=alert(1)>`	Escaped in the cart, grid, receipt, and history; never executes	S1
PS-124	SEC	Transitions need the session token	(Dev) POST a cart/pay transition without the token	Rejected (offline export/sync are deliberately token-free — verify those are still org-scoped)	S2

13. Cross-flow / regression & the double-engine watch

ID	Type	Title	Expected result	Sev
XF-001	FUNC	**Single cart engine**	The cart updates **once** per action — no flicker/double-line/double-POST. (There is a legacy inline engine that must stand down to the external `pos-terminal.js`.) Watch the Network tab for duplicate `ajaxCart` posts	S1
XF-002	GST	POS bill ↔ Sales invoice ↔ Tax engine	The bill's GST, the synced invoice's GST, and what the Tax engine (Part 6) computes all agree	S1
XF-003	LIFE	Full day	Open → several sales (cash/card/upi/split/credit/hold/recall/amend) → close; the session cash reconciles and bill numbers are gap-free	S1
XF-004	ISO	Two-tenant	Nothing tenant A rings up is visible to tenant B	S1

```
Defect ID:        ZE-POS-____
Linked Case ID:   PS-0__ / ISO-0__ / XF-0__
Title / Severity / Type:
Environment:      Prod | <browser + scanner/printer> | tenant: <org> · terminal: <id> (NEVER MARUTHI)
Steps + amounts:  cart lines, mode, tax on/off, payment method(s) + amounts
Expected (by hand): taxable, CGST/SGST/IGST, round-off, payable, change, bill no.
Actual:           (same) — attach screenshot + the receipt + Console/Network
Session/bill ids:
```

14. Areas to probe hardest (attention map)

1. **Payment integrity (PS-043/044/045)** — split must sum to ≥ payable; no double-charge on retry/double-click; change is exact. This is the money.
2. **Bill numbering under concurrency (PS-086)** — two terminals at once must never duplicate or skip a number; per org+FY.
3. **GST split on the bill *and* the synced invoice (PS-030/031/091, XF-002)** — CGST+SGST intra vs IGST inter, and the invoice matches the bill and the tax engine.
4. **The cart double-engine (XF-001)** — verify only the external `pos-terminal.js` runs (the legacy inline engine stands down); watch for double `ajaxCart` posts / flicker / double lines.
5. **Qty edit never deletes (PS-007)** and **unpriced-product-in-fixed-mode block (PS-022)** — common till mistakes.
6. **Hold/Recall/Void carry-forward (PS-060..064)** — after hold/void the next add must start a **fresh** cart (not target the held/voided one); recall auto-holds the current cart.
7. **Amend an issued bill (PS-070)** — same number, stock delta, reconciled refund/balance, audit row — never a silent renumber or a stock double-count.
8. **Receipt never undoes a completed sale (PS-052)** — a print failure must leave the sale completed.

9. **Tax default state on load (PS-024) — fixed (80dd4b0).** The UI now defaults to the same OFF state as the billing logic, so a fresh session shows "TAX: OFF" honestly; re-verify the button matches the computed tax and that the toggle works both ways.
10. **Multi-tenant isolation (§11)** — especially the child entities (payments/items/held/mode-price) that rely on their parent's org scope; and that offline export/sync (token-free) stay org-scoped.

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15. Teardown (mandatory)

1. **List every test artefact** — sessions (close them), bills, held bills, credit bills, amendments, terminals, products, receipt templates, offline queue — with ids.
2. **Close any open test session** and discard leftover PtCreated carts; don't leave an open session on a terminal.
3. **Void/delete** test bills (or hand the dev the list); never touch a real bill's number.
4. Delete test products/terminals/templates and any throwaway tenant (see the Auth/Signup teardown recipe).
5. Confirm you **never touched MARUTHI** (org `100000`) or any live tenant's terminal, sessions, bills, or payments.
6. Attach the run sheet(s) + defect ids to this part's report.

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16. Known dead / fragile spots (for the tester's awareness)

- `EditTransaction.xml` (the `/apps/zenith/pos/PosTransactions → EditTransaction` back-office bill screen) **works** — a re-check corrected the earlier recon: the GST fields (`customerGstin` , `isB2b` , `cgstTotal/sgstTotal/igstTotal`) are declared on `PosTransaction` via the `zenith-erp-india` extension (`IndiaExtendEntities.xml`), and `generate#UpiQrString` exists in that same component (`IndiaPosServices.xml`); the screen renders HTTP 200 and derives the GST split for display. It is a legacy back-office **bill viewer / draft-cart editor** (hidden from the menu); the primary live bill-edit path is still the Terminal's amend flow (PS-070), but this screen is not broken.
- `.bak` **files** (`Terminal.xml.bak` , `ReceiptPdf.xml.bak` , `PosServices.xml.bak`) are stale copies, not deployed — ignore.
- **/qapps** strips `<script>` / `<style>` — the rich terminal only works on **/apps** (verify the /qapps→/apps bounce, PS-121).

End of Part 9 — POS. Related: Part 6 — Tax (the GST engine the till uses); Part 4 — Sales (the invoice the bill syncs to); Part 7 — Barcode (scan resolution); Part 2 — Contacts (the customer attached). The highest-risk areas to raise with the dev after a run: any duplicate/skipped bill number under concurrency, any double-charge or split-total mismatch, any GST mismatch between the bill and its synced invoice, and any cart double-engine flicker/double-post.